MTH 418a: Inference-I
Assignment No. 4: Basu's Theorem

1. Let $X_1, \ldots, X_n$ be a random sample from $N(\mu, \sigma^2)$, $\mu \in \mathbb{R}, \sigma > 0$.

   (i) Show that $\sum_{i=1}^n (X_i - \mu)^2$ and $U = \frac{\overline{X} - \mu}{\sqrt{\sum_{i=1}^n (X_i - \mu)^2}}$ are independent. Are $\sum_{i=1}^n (X_i - \mu)^2$ and $V = \frac{\overline{X} - \mu}{\sqrt{\sum_{i=1}^n (X_i - \overline{X})^2}}$ also independent? Find the mean and variance of U^2.

   (ii) Show that $\overline{X}$ and $\underline{T} = (X_1 - \overline{X}, \ldots, X_n - \overline{X})$ are independent. In particular, show that $\overline{X}$ and $W = \sum_{i=1}^n (X_i - \overline{X})^2$ are independent.

   (iii) Find the conditional distribution function, conditional density and conditional expectation of of X_1, given $\overline{X} = t, t \in \mathbb{R}$.

   (iv) Using the m.g.f. and findings in (ii), find the distribution of $W = \sum_{i=1}^n (X_i - \overline{X})^2$.

2. Let $X_1, \ldots, X_n$ be a random sample from $Exp(\mu, \sigma)$, $\mu \in \mathbb{R}, \sigma > 0$.

   (i) Show that $\sum_{i=1}^n (X_i - \mu)$ and $U = \frac{X_{(1)} - \mu}{\sum_{i=1}^n (X_i - \mu)}$ are independent. Are $\sum_{i=1}^n (X_i - \mu)$ and $V = \frac{X_{(1)} - \mu}{\sum_{i=1}^n (X_i - X_{(1)})}$ also independent? Find the mean and variance of U.

   (ii) Show that $X_{(1)}$ and $\underline{T} = (X_1 - X_{(1)}, \ldots, X_n - X_{(n)})$ are independent. In particular, show that $X_{(1)}$ and $\sum_{i=1}^n (X_i - X_{(1)})$ are independent.

   (iii) Find the conditional distribution function, conditional density and conditional expectation of X_1, given $X_{(1)} = t, t > \mu$.

   (iv) Using the m.g.f. and findings in (ii), find the distribution of $W = \sum_{i=1}^n (X_i - X_{(1)})$.

   (v) Let $Z_i = \frac{X_{(n)} - X_{(i)}}{X_{(n)} - X_{(n-1)}}, i = 1, \ldots, n-2$. Show that $\underline{T} = (X_{(1)}, \sum_{i=1}^n (X_i - X_{(1)}))$ and $\underline{Z} = (Z_1, \ldots, Z_{n-2})$ are independent.

3. Let $X_1, \ldots, X_n$ be a random sample from $U(\theta_1, \theta_2)$, $-\infty < \theta_1 < \theta_2 < \infty$. Let $Z_i = \frac{X_{(i)} - X_{(1)}}{X_{(n)} - X_{(1)}}, i = 2, \ldots, n-1$. Show that $\underline{T} = (X_{(1)}, X_{(n)})$ and $\underline{Z} = (Z_2, \ldots, Z_{n-1})$ are independent.

4. Let $X_1, \ldots, X_n$ be a random sample from $U(0, \theta)$, $\theta > 0$.

   (i) Show that $X_{(n)}$ and $\underline{T} = (\frac{X_1}{X_{(n)}}, \ldots, \frac{X_n}{X_{(n)}})$ are independent.

   (ii) Find $E(\frac{X_1}{X_{(n)}})$. Also find the conditional distribution function, conditional density and conditional expectation of X_1, given $X_{(n)} = t$, $0 < t < \theta$.

   (iii) Find $E(\frac{X_{(r)}}{X_{(n)}}), r = 1, \ldots, n-1$.

5. Let $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_n$ be independent random samples from $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$, respectively, $-\infty < \mu_i < \infty, \sigma_i > 0, i = 1, 2$. Show that $\underline{T} = (\overline{X}, \overline{Y}, \sum_{i=1}^n (X_i - \overline{X})^2, \sum_{i=1}^n (Y_i - \overline{Y})^2)$ and

$$V(\underline{X}) = \frac{\sum_{i=1}^n (X_i - \overline{X})(Y_1 - \overline{Y})}{\sqrt{\sum_{i=1}^n (X_i - \overline{X})^2 \sum_{i=1}^n (Y_i - \overline{Y})^2}}$$

are statistically independent.

MTH 418a: Inference-I

Assignment No.4: Basu's Theorem

Solutions

$\boxed{\text{Problem No.1}}$ (i) Let us fix $\mu \in \mathbb{R}$ and consider $\sigma \in (0, \infty)$ as the parameter. Then $\sum_{i=1}^{n}(X_i-\mu)^2$ is a complete-sufficient statistic for σ.

Let $Z_i = \frac{X_i-\mu}{\sigma}$, $i=1,\dots,n$. Then $Z_1,\dots,Z_n$ are i.i.d. $N(0,1)$ so that the distribution of $\underline{Z}=(Z_1,\dots,Z_n)$ does not depend on parameter σ; $X_i = \mu+\sigma Z_i$ $i=1,\dots,n$.

$$U = \frac{\bar{X}-\mu}{\sqrt{\sum_{i=1}^{n}(X_i-\mu)^2}} = \frac{\bar{Z}}{\sqrt{\sum_{i=1}^{n} Z_i^2}} \text{ is ancillary}$$

$$V = \frac{\bar{X}-\mu}{\sqrt{\sum_{i=1}^{n}(X_i-\bar{X})^2}} = \frac{\bar{Z}}{\sqrt{\sum_{i=1}^{n}(Z_i-\bar{Z})^2}} \text{ is ancillary}$$

By Basu's Theorem $T=\sum_{i=1}^{n}(X_i-\mu)^2$ and U are independent

$T=\sum_{i=1}^{n}(X_i-\mu)^2$ and V are independent

$$E(TU^2) = E(T)\,E(U^2)$$

$$E((\bar{X}-\mu)^2) = E\left(\sum_{i=1}^{n}(X_i-\mu)^2\right) E(U^2)$$

$$E(U^2) = \frac{V(\bar{X})}{E(\sigma^2 \chi_n^2)} = \frac{\sigma^2/n}{n\sigma^2} = \frac{1}{n^2}$$

$$E(T^2U^4) = E(T^2)\,E(U^4)$$

$$E((\bar{X}-\mu)^4) = E\left(\left(\sum_{i=1}^{n}(X_i-\mu)^2\right)^2\right) E(U^4)$$

$$E(U^4) = \frac{E((\bar{X}-\mu)^4)}{E((\sigma^2\chi_n^2)^2)} = \frac{\frac{n^2}{\sigma^4}\times 3}{\sigma^4[2n+n^2]} = \frac{3n}{n+2}$$

$$V(U^2) = E(U^4)-(E(U^2))^2 = \frac{3n}{n+2}-\frac{1}{n^4} = \frac{3n^5-n-2}{n^4(n+2)}$$

(ii) Fix $\sigma>0$ and consider $\mu\in\mathbb{R}$ as the parameter. Then $\bar{X}$ is a complete-sufficient statistic for μ. Let $Z_i = X_i-\mu$, $i=1,\dots,n$. Then $Z_1,\dots,Z_n$ are iid $N(0,1)$, so that the distribution of $\underline{Z}=(Z_1,\dots,Z_n)$ does not depend on the parameter μ

$$\Rightarrow \underline{T}=(X_1-\bar{X},\dots,X_n-\bar{X})=(Z_1-\bar{Z},\dots,Z_n-\bar{Z}) \text{ is ancillary}$$

By Basu's Theorem $\bar{X}$ and $\underline{T}=(X_1-\bar{X},\dots,X_n-\bar{X})$ are independent

$$\Rightarrow \bar{X} \text{ and } W=\sum_{i=1}^{n}(X_i-\bar{X})^2 \text{ are independent.}$$

(iii) Fix $t\in\mathbb{R}$. Then, for $x\in\mathbb{R}$,

$$F_{X_1|\bar{X}}(x|t) = P(X_1\le x\mid \bar{X}=t)$$

$$= P(X_1-\bar{X}\le x-t\mid \bar{X}=t)$$

$$= P(X_1-\bar{X}\le x-t) \quad (\text{by (ii) } \bar{X} \text{ and } X_1-\bar{X} \text{ are independent})$$

$$= \Phi\left(\sqrt{\frac{n}{n-1}}\,\frac{(x-t)}{\sigma}\right) \quad \left(X_1-\bar{X}\sim N\left(0,\frac{n-1}{n}\sigma^2\right)\right)$$

$$f_{X_1|\bar{X}}(x|t) = \sqrt{\frac{n}{n-1}}\cdot\frac{1}{\sigma}\,\phi\left(\sqrt{\frac{n}{n-1}}\,\frac{(x-t)}{\sigma}\right), \quad -\infty<x<\infty$$

$$\rightarrow \text{p.d.f. of } N\left(t, \frac{(n-1)\sigma^2}{n}\right)$$

$$E(X_1\mid\bar{X}=t)=t$$

<u>Aliter</u> For fixed $t\in\mathbb{R}$

$$E(X_1\mid\bar{X}=t) = E(X_1-\bar{X}+t\mid\bar{X}=t)$$

$$= t+E(X_1-\bar{X}\mid\bar{X}=t)$$

$$= t+E(X_1-\bar{X}) \quad (\text{by (ii) } \bar{X} \text{ and } X_1-\bar{X} \text{ are independent})$$

$$= t$$

(iv) $$\sum_{i=1}^{n}(x_i-\mu)^2 = W + n(\bar{x}-\mu)^2$$

$$T_1 = W + U,$$

W and U are independently distributed

$$T_1 \sim \sigma^2 \chi^2_n, \quad U \sim \sigma^2 \chi^2_1$$

$$M_{T_1}(t) = M_{W+U}(t)$$

$$= M_W(t)\, M_U(t)$$

$$(1-2\sigma^2 t)^{-\frac{n}{2}} = M_W(t)\,(1-2\sigma^2 t)^{-\frac{1}{2}} \qquad t<\frac{1}{2\sigma^2}$$

$$\Rightarrow \quad M_W(t) = (1-2\sigma^2 t)^{-\frac{n-1}{2}}, \qquad t<\frac{1}{2\sigma^2}$$

$\hookrightarrow$ m.g.f. of $\sigma^2\chi^2_{n-1}$

By the uniqueness of the m.g.f. $W \sim \sigma^2 \chi^2_{n-1}$

$\boxed{\text{Problem No. 2}}$ (i) Fix $\mu \in \mathbb{R}$ and consider $\sigma>0$ as the parameter. Then $\sum_{i=1}^{n}(X_i-\mu)$ is a complete-sufficient statistic. Define $Z_i = \frac{X_i-\mu}{\sigma}$, $i=1,\dots,n$, so that $Z_1,\dots,Z_n$ are i.i.d. $Exp(1)$, i.e. the distribution of $\underline{Z}=(Z_1,\dots,Z_n)$ does not depend on the parameter σ.

$$\Rightarrow \quad U = \frac{X_{(1)}-\mu}{\sum_{i=1}^{n}(X_i-\mu)} = \frac{Z_{(1)}}{\sum_{i=1}^{n} Z_i} \text{ is ancillary}$$

By Basu's Theorem, $T=\sum_{i=1}^{n}(X_i-\mu)$ and U are independent

$$E(TU) = E(T)\,E(U)$$

$$E(X_{(1)}-\mu) = E(T)\,E(U)$$

$$\frac{n(X_{(1)}-\mu)}{\sigma} \sim Exp(1), \quad \frac{T}{\sigma} \sim Gamma(1, n)$$

$$\Rightarrow \quad E(U) = \frac{1/n}{n} = \frac{1}{n^2};$$

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$$E(T^2U^2) = E(T^2)\, E(U^2)$$

$$E((X_{(1)}-\mu)^2) = E(T^2)\, E(U^2)$$

$$E(U^2) = \frac{2\sigma^2/n^2}{n(n+1)\sigma^2} = \frac{2}{n^3(n+1)}$$

$$V(U) = \frac{2}{n^3(n+1)} - \frac{1}{n^4} = \frac{2n-1}{n^4(n+1)}.$$

(ii) Fix $\sigma>0$ and consider $\mu \in \mathbb{R}$ as the parameter. Then $X_{(1)}$ is a complete-sufficient statistic for $\mu \in \mathbb{R}$. Let $Z_i = X_i - \mu$, $i=1..n$, so that $Z_1, \dots, Z_n$ are iid $Exp(0, \sigma)$ i.e. the distribution of $\underline{Z} = (Z_1, \dots, Z_n)$ does not depend on the unknown parameter μ.

Distribution of

$$\underline{I} = (X_1 - X_{(1)}, \dots, X_n - X_{(1)}) = (Z_1 - Z_{(1)}, \dots, Z_n - Z_{(1)})$$

does not depend on the parameter μ i.e. $\underline{I}$ is ancillary.

By Basu's Theorem

$X_{(1)}$ and $\underline{I} = (X_1 - X_{(1)}, \dots, X_n - X_{(1)})$ are independent

$\Rightarrow$ $X_{(1)}$ and $W = \sum_{i=1}^{n} (X_i - X_{(1)})$ are independent

(iii) Fix $t \in \mathbb{R}$. Then, for $x \in \mathbb{R}$,

$$F_{X_1|T}(x|t) = P(X_1 \le x \mid X_{(1)} = t)$$

$$= P(X_1 - X_{(1)} \le x - t \mid X_{(1)} = t)$$

$$= P(X_1 - X_{(1)} \le x - t) \quad \text{(by (ii) } X_1 - X_{(1)} \text{ and } X_{(1)} \text{ are independent)}$$

For $x < t$, $F_{X_1|T}(x|t) = 0$. For $x \ge t$

$$F_{X_1|T}(x|t) = P(X_1 - X_{(1)} \le x - t)$$

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$= P(X_{(1)} \geq X_1 - (x-t))$

$= P(X_j > X_1 - (x-t), \; j=1,\dots,n)$

$= P(X_j > X_1 - (x-t), \; j=2,\dots,n)$

$= P(Z_j > Z_1 - (x-t), \; j=2,\dots,n)$ ($Z_i = X_i - \mu$, $i=1,\dots,n$, so that $Z_1,\dots,Z_n$ are iid $Exp(0,\sigma)$)

$$= \int_0^\infty P(Z_j > z-(x-t), \; j=2,\dots,n \mid Z_1=z)\, \frac{1}{\sigma} e^{-z/\sigma}\, dz$$

$$= \int_0^\infty P(Z_j > z-(x-t), \; j=2,\dots,n)\, \frac{1}{\sigma} e^{-z/\sigma}\, dz$$

$$= \int_0^\infty \left[P(Z_1 > z-(x-t)) \right]^{n-1} \frac{1}{\sigma} e^{-\frac{z}{\sigma}}\, dz$$

$$= \int_0^{x-t} \frac{1}{\sigma} e^{-\frac{z}{\sigma}}\, dz + \int_{x-t}^\infty e^{-\frac{(n-1)(z-(x-t))}{\sigma}} \frac{1}{\sigma} e^{-\frac{z}{\sigma}}\, dz$$

$$= 1 - e^{-\frac{(x-t)}{\sigma}} + e^{\frac{(n-1)(x-t)}{\sigma}} \times \frac{1}{n} e^{-\frac{n(x-t)}{\sigma}}$$

$$= 1 - \frac{n-1}{n} e^{-\frac{(x-t)}{\sigma}}$$

Thus, for any fixed $t \in \mathbb{R}$,

$$F_{X_1 | X_{(1)}}(x|t) = \begin{cases} 0, & x < t \\ 1 - \frac{n-1}{n} e^{-\frac{(x-t)}{\sigma}}, & x \geq t \end{cases}$$

$\rightarrow$ has a jump of size $\frac{1}{n}$ at $x=t$.

$$f_{X_1 | X_{(1)}}(x|t) = \begin{cases} \frac{1}{n}, & x=t \\ \frac{n-1}{n\sigma} e^{-\frac{(x-t)}{\sigma}}, & x > t \end{cases}$$

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$$E(X_1 \mid X_{(1)}=t) = \frac{1}{n}\,x + \int_t^{\infty} \lambda \times \frac{n-1}{n\sigma} e^{-\frac{(\lambda-t)}{\sigma}}\,d\lambda$$

$$= \frac{t}{n} + \frac{n-1}{n}(t+\sigma)$$

$$= t + \frac{n-1}{n}\sigma$$

<u>Altern</u> $\quad E(X_1 \mid X_{(1)}=t) = E(X_1 - X_{(1)} + t \mid X_{(1)}=t)$

$$= E(X_1 - X_{(1)} \mid X_{(1)}=t) + t$$

$$= E(X_1 - X_{(1)}) + t \quad (\text{by (ii) } X_1 - X_{(1)} \text{ and } X_{(1)} \text{ are independent})$$

$$= t + [\mu + \sigma - (\mu + \tfrac{\sigma}{n})] \quad (X_{(1)} \sim Exp(\mu, \tfrac{\sigma}{n}))$$

$$= t + \frac{n-1}{n}\sigma$$

(IV) $\quad \sum_{i=1}^{n}(X_i - \mu) = \sum_{i=1}^{n}(X_i - X_{(1)}) + n(X_{(1)} - \mu)$

$$T_1 = W + U$$

Where

$$T_1 = \sum_{i=1}^{n}(X_i - \mu) \sim \text{Gamma}(\sigma, n)$$

$$W = \sum_{i=1}^{n}(X_i - X_{(1)})$$

$$U = n(X_{(1)} - \mu) \sim Exp(0, \sigma)$$

W and U independent (by (ii))

$$M_{T_1}(t) = M_{W+U}(t)$$

$$= M_W(t)\, M_U(t)$$

$$(1-\sigma t)^{-n} = M_W(t)\,(1-\sigma t)^{-1}, \quad t < \frac{1}{\sigma}$$

$$\Rightarrow \quad M_W(t) = (1-\sigma t)^{-(n-1)}, \quad t < \frac{1}{\sigma}$$

$\hookrightarrow$ m.g.f. of Gamma $(\sigma, n-1)$

By the uniqueness of the m.g.f. $W \sim$ Gamma $(\sigma, n-1)$

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(V) Consider $\underline{\theta} = (\mu, \sigma) \in \mathbb{R} \times (0, \infty) = \Theta$, say, as the parameter. Then $\underline{T} = (X_{(1)}, \sum_{i=1}^{n} (X_i - X_{(1)}))$ is a Complete-Sufficient statistic for $\underline{\theta}$. Let $Y_i = \frac{X_i - \mu}{\sigma}$, $i=1,\dots,n$. Then $Y_1, \dots, Y_n$ are iid $\mathrm{Exp}(0, 1)$, i.e. the distribution of $\underline{Y} = (Y_1, \dots, Y_n)$ does not depend on the parameter $\underline{\theta}$. We have $X_i = \mu + \sigma Y_i$, $i=1,\dots,n$, $X_{(i)} = \mu + \sigma Y_{(i)}$, $i=1,\dots,n$.

The distribution of

$$(Z_1, \dots, Z_{n-2}) = \left(\frac{Y_{(n)} - Y_{(1)}}{Y_{(n)} - Y_{(n-1)}}, \dots, \frac{Y_{(n)} - Y_{(n-2)}}{Y_{(n)} - Y_{(n-1)}} \right)$$ does not depend

on parameter $\underline{\theta}$, i.e. $\underline{Z} = (Z_1, \dots, Z_{n-2})$ is ancillary.

By Basu's Theorem, $\underline{T}$ and $\underline{Z}$ are independent.

$\boxed{\text{Problem No. 3}}$ $\underline{T} = (X_{(1)}, X_{(n)})$ is Complete-Sufficient for $\underline{\theta} = (\theta_1, \theta_2)$.

Let $Y_i = \frac{X_i - \theta_1}{\theta_2 - \theta_1}$, $i=1,\dots,n$. The $Y_1, \dots, Y_n$ are iid $U(0, 1)$, i.e. the distribution of $(Y_1, \dots, Y_n)$ does not depend on unknown parameter $\underline{\theta}$, $X_i = \theta_1 + (\theta_2 - \theta_1) Y_i$, $i=1,\dots,n$, $X_{(i)} = \theta_1 + (\theta_2 - \theta_1) Y_{(i)}$, $i=1,\dots,n$. The distribution of

$$(Z_2, \dots, Z_{n-1}) = \left(\frac{X_{(2)} - X_{(1)}}{X_{(n)} - X_{(1)}}, \dots, \frac{X_{(n-1)} - X_{(1)}}{X_{(n)} - X_{(1)}} \right)$$

$$= \left(\frac{Y_{(2)} - Y_{(1)}}{Y_{(n)} - Y_{(1)}}, \dots, \frac{Y_{(n-1)} - Y_{(1)}}{Y_{(n)} - Y_{(1)}} \right)$$

does not depend on unknown parameter $\underline{\theta}$, i.e. $\underline{Z} = (Z_2, \dots, Z_{n-1})$ is ancillary.

By Basu's Theorem $\underline{T} = (X_{(1)}, X_{(n)})$ and $\underline{Z}$ are ancillary.

Problem No. 4

(i) $U = X_{(n)}$ is a complete-sufficient statistic for θ.

Let $Z_i = \frac{X_i}{\theta}$, $i=1,\dots,n$. Then $Z_1,\dots,Z_n$ are iid $U(0,1)$, i.e. the distribution of $\underline{Z} = (Z_1,\dots,Z_n)$ does not depend on unknown parameter θ, $X_i = \theta Z_i$, $i=1,\dots,n$, $X_{(i)} = \theta Z_{(i)}$, $i=1,\dots,n$. The distribution of

$$\underline{T} = \left(\frac{X_1}{X_{(n)}},\dots,\frac{X_n}{X_{(n)}}\right) = \left(\frac{Z_1}{Z_{(n)}},\dots,\frac{Z_n}{Z_{(n)}}\right)$$

does not depend on the parameter θ, i.e. $\underline{T}$ is ancillary

By the Basu Theorem $U = X_{(n)}$ and $\underline{T}$ are independent.

(ii) $E(X_1) = E\left(\frac{X_1}{X_{(n)}} X_{(n)}\right) = E\left(\frac{X_1}{X_{(n)}}\right) E(X_{(n)})$ (by (i) $\frac{X_1}{X_{(n)}}$ and $X_{(n)}$ are independent)

$$E\left(\frac{X_1}{X_{(n)}}\right) = \frac{E(X_1)}{E(X_{(n)})} = \frac{E(Z_1)}{E(Z_{(n)})} = \frac{1/2}{n/n+1} = \frac{n+1}{2n}$$

For fixed $t \in (0,\theta)$,

$$F_{X_1|X_{(n)}}(x|t) = P(X_1 \le x \mid X_{(n)} = t)$$

$$= P_\theta\left(\frac{X_1}{X_{(n)}} \le \frac{x}{t} \,\Big|\, X_{(n)} = t\right)$$

$$= P_\theta\left(\frac{X_1}{X_{(n)}} \le \frac{x}{t}\right) \quad \left(\text{By (i) } \frac{X_1}{X_{(n)}} \text{ and } X_{(n)} \text{ are independent}\right)$$

$$= P_\theta\left(\frac{Z_1}{Z_{(n)}} \le \frac{x}{t}\right)$$

For $x<0$, $F_{X_1|X_{(n)}}(x|t) = 0$ and, for $0 \le x < t$,

$$F_{X_1|X_{(n)}}(x|t) = P\left(\frac{Z_1}{Z_{(n)}} \le \frac{x}{t}\right)$$

$$= 1 - P\left(Z_{(n)} \le \frac{t}{x} Z_1\right)$$

$$= 1 - P\left(Z_j \le \frac{t}{x} Z_1,\ j=1,\dots,n\right)$$

$$= 1 - P\left(Z_j \le \frac{t}{x} Z_1,\ j=2,\dots,n\right)$$

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$$= 1 - \int_0^1 P(Z_j \le \tfrac{t}{x} z,\ j = 2 \cdots, n \mid Z_1 = z)\, dz$$

$$= 1 - \int_0^1 P(Z_j \le \tfrac{t}{x} z,\ j = 2 \cdots, n)\, dz$$

$$= 1 - \int_0^1 [P(Z_1 \le \tfrac{t}{x} z, 1)]^{n-1} dz$$

$$= 1 - \left[\left(\tfrac{t}{x}\right)^{n-1} \int_0^{x/t} z^{n-1} dz + \int_{x/t}^1 dz\right]$$

$$= 1 - \left[\left(\tfrac{t}{x}\right)^{n-1} \times \tfrac{1}{n} \times \left(\tfrac{x}{t}\right)^n + 1 - \tfrac{x}{t}\right]$$

$$= \frac{n-1}{n} \frac{x}{t}$$

Also, for $x \ge t$, $F_{X_1|X_{(n)}}(x|t) = 1$. Thus

$$F_{X_1|X_{(n)}}(x|t) = \begin{cases} 0, & x < 0 \\ \frac{n-1}{n}\frac{x}{t}, & 0 \le x < t \\ 1, & x \ge t \end{cases}$$

$\longrightarrow$ has a jump of size $\frac{1}{n}$ at $x = t$

$$f_{X_1|X_{(n)}}(x|t) = \begin{cases} \frac{n-1}{n} \times \frac{1}{t}, & 0 \le x < t \\ \frac{1}{n}, & x = t \\ 0, & \text{otherwise} \end{cases}$$

$$E(X_1 | X_{(n)} = t) = \int_0^t x \times \frac{n-1}{nt} dx + t \times \frac{1}{n} = \frac{n-1}{2n} t + \frac{t}{n}$$

$$= \frac{n+1}{2n} t$$

<u>Alter.</u> $E(X_1 | X_{(n)} = t) = E\left(\frac{X_1}{X_{(n)}} \times t \,\middle|\, X_{(n)} = t\right) = t\, E\left(\frac{X_1}{X_{(n)}} \,\middle|\, X_{(n)} = t\right)$

$= t\, E\left(\frac{X_1}{X_{(n)}}\right)$ ($\frac{X_1}{X_{(n)}}$ are $X_{(n)}$ are independent by (i))

$\boxed{9/10}$ $= \frac{n+1}{2n} t$ (by (ii))

(iv) $E(X_{(r)}) = E\left(\frac{X_{(r)}}{X_{(n)}} \times X_{(n)} \right)$

$= E\left(\frac{X_{(r)}}{X_{(n)}} \right) E(X_{(n)})$ $\left(\text{by (i) } X_{(n)} \text{ and } \frac{X_{(r)}}{X_{(n)}} \text{ are independent} \right)$

$$E\left(\frac{X_{(r)}}{X_{(n)}} \right) = \frac{E(X_{(r)})}{E(X_{(n)})} = \frac{E(Z_{(r)})}{E(Z_{(n)})} = \frac{\frac{\lfloor n}{\lfloor r-1 \lfloor n-r} \int_0^1 z \times z^{r-1}(1-z)^{n-r}\,dz}{\frac{\lfloor n}{\lfloor n-1 \lfloor 0} \int_0^1 z \times z^{n-1}(1-z)^0\,dz}$$

$= \frac{r}{n}$, $r = 1, \ldots, n$

$\boxed{\text{Problem No. 5}}$ The joint pdf of $(X_1, \ldots, X_n, Y_1, \ldots, Y_n)$ is

$$f_{\underline{X}, \underline{Y}}(\underline{x}, \underline{y}) = e^{\frac{n\mu_1}{\sigma_1^2}\bar{x} + \frac{n\mu_2}{\sigma_2^2}\bar{y} - \frac{1}{2\sigma_1^2}\sum x_i^2 - \frac{1}{2\sigma_2^2}\sum y_i^2 - B(\underline{\mu}, \sigma^2)},$$

$(\eta_1, \eta_2, \eta_3, \eta_4) = \left(\frac{n\mu_1}{\sigma_1^2}, \frac{n\mu_2}{\sigma_2^2}, -\frac{1}{2\sigma_1^2}, \frac{-1}{2\sigma_2^2} \right) \in \mathbb{R}^2 \times (-\infty, 0)^2$ contains an open rectangle in $\mathbb{R}^4$

$\Rightarrow$ $\underline{U} = (\bar{X}, \bar{Y}, \sum_{i=1}^n X_i^2, \sum_{i=1}^n Y_i^2)$ is a complete sufficient statistic for $\underline{\theta} = (\mu_1, \mu_2, \sigma_1^2, \sigma_2^2)$

$\Rightarrow$ $\underline{T} = (\bar{X}, \bar{Y}, \sum_{i=1}^n (X_i - \bar{X})^2, \sum_{i=1}^n (Y_i - \bar{Y})^2)$, being a 1-1 function of complete-sufficient statistic $\underline{U}$, is complete-sufficient.

Let $S_i = \frac{X_i - \mu_1}{\sigma_1}$, $i = 1, \ldots, n$, $V_i = \frac{Y_i - \mu_2}{\sigma_2}$, $i = 1, \ldots, n$. Then $S_1, \ldots, S_n, V_1, \ldots, V_n$ are iid $N(0, 1)$, i.e. the distribution of $(S_1, \ldots, S_n, V_1, \ldots, V_n)$ does not depend on parameter $\underline{\theta}$. $X_i = \mu_1 + \sigma_1 S_i$, $Y_i = \mu_2 + \sigma_2 V_i$, $i = 1, \ldots, n$, $\bar{X} = \mu_1 + \sigma_1 \bar{S}$, $\bar{Y} = \mu_2 + \sigma_2 \bar{V}$ and distribution of

$$V(\underline{X}) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 \sum_{i=1}^n (Y_i - \bar{Y})^2}} = \frac{\sum (S_i - \bar{S})(V_i - \bar{V})}{\sqrt{\sum (S_i - \bar{S})^2 \sum (V_i - \bar{V})^2}}$$

does not depend on parameter $\underline{\theta}$, i.e. V is ancillary

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$\Rightarrow$ $\underline{T}$ and $\underline{V}$ are independent (Basu's Thm)